

AGENTIC AI · HEALTHCARE INTEGRATION OPERATIONS

Interface System Problem AI Resolver

An autonomous diagnostic & remediation agent for Mirth Connect / NextGen Connect — it investigates the live engine, diagnoses the incident, and hands the on-call team a fix.

Prepared for Senior Management & CXO Review

Functional validation · Business impact · Customer value

The one-minute version

02

Healthcare runs on interface engines. When a Mirth channel silently stops or a queue backs up, patient data stops flowing — and today that is caught late, by scarce specialists, under pressure.

The problem

Mirth fails in ~50 well-known ways across 11 layers.
Detection is manual, expertise is scarce, and
mean-time-to-resolve runs into hours.

What we built

An agentic AI that senses the engine, reasons over an
11-layer failure taxonomy,
and produces an evidence-based diagnosis +
prioritized runbook — with safe, approved actions.

The payoff

Faster resolution, fewer outages, and senior expertise
on every on-call shift —
deployable even in air-gapped, PHI-sensitive hospital
networks.

11

failure layers
continuously validated

3

engines in one tool:
LLM · rule-based · RAG

79

automated tests
proving the behavior

0

extra dependencies &
no PHI leaves the host

Interface downtime is a clinical and business risk

Mirth Connect is the backbone moving HL7/FHIR messages between EHRs, labs, pharmacies and devices. Its failure modes are known — but operationally hard to catch and resolve.

Silent failures

A channel stops or drops messages with no alert. Data flow halts before anyone notices.

Scarce expertise

Root-causing OOM, Derby corruption or queue backpressure needs senior engineers.

Slow MTTR

Triage is manual: hunt logs, check queues, read stats, correlate — often at 2 a.m.

Monitoring blind spots

Stopped channels, rising errors, full disks — often unmonitored until impact.

The 5 dominant real-incident patterns (what actually takes systems down)

Memory exhaustion from an unpruned message store · Derby run in production · Unmonitored destination queues growing while an endpoint is down
Transformer exceptions on non-conformant messages (often silent) · Missing alerting — a stopped channel, rising errors or a full disk with nobody notified

An agent that works the case like a senior SRE

Give it a symptom — “queue climbing”, “engine frozen” — and it runs a disciplined investigation loop, end to end, in seconds.

1 GATHER

Pull channel states, queues, engine/host stats, message-store size & logs

2 CORRELATE

Reconstruct the causal chain — trigger → amplifier → observed failure

3 DIAGNOSE

Classify against the 11-layer taxonomy; quantify every finding

4 REMEDIATE

Prioritized runbook: stop-the-bleeding now + durable root-cause fix

Every diagnosis is evidence-based and structured

Grounded, not guessed

Cites the actual numbers (heap 95%, queue 47,113). States what it ruled out, with a confidence score.

Actionable

Immediate + durable steps, each with a way to verify it worked, plus the monitoring to prevent recurrence.

Safe by design

Only whitelisted safe actions are offered — and only with explicit human approval.

Architecture at a glance

SENSES (read-only)

Channel states
Destination queues
Engine / host stats
Mirth log tail
Message-store size & pruning

REASONING ENGINE

Investigation loop with tool-use

Grounded in the 11-layer failure taxonomy

Choose your engine:

- LLM agent (Claude) — deepest reasoning
- Local rule-based — offline, deterministic
- RAG case-memory — learns from history

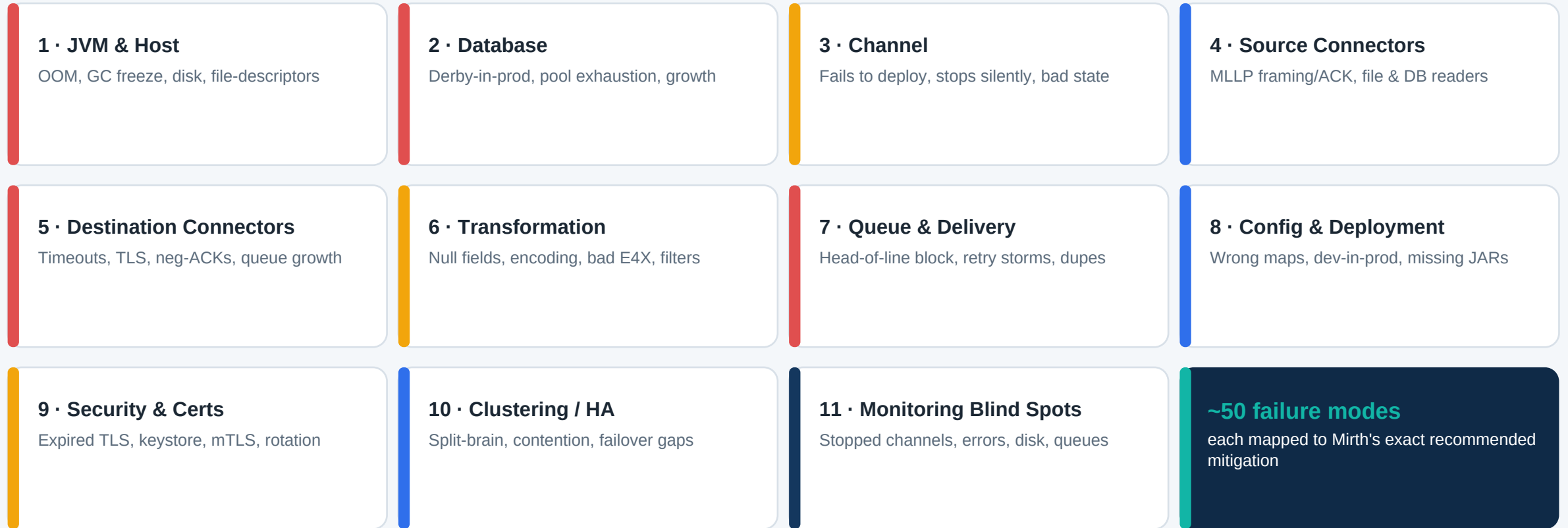
STRUCTURED OUTPUT

Severity + confidence
Key signals & causal chain
Layer classification
Immediate + durable runbook
Safe suggested actions

Human-in-the-loop remediation

Whitelisted safe actions (start channel, redeploy channel) execute only after explicit confirmation. Anything destructive or config-changing is recommend-only — written into the runbook for a human to perform.

The Mirth areas we validate — an 11-layer failure taxonomy



Every diagnosis is anchored to this taxonomy — so guidance is consistent, explainable, and vendor-aligned.

One incident, worked end-to-end

Symptom reported: “engine frozen, OOM around 02:10.” The agent gathers evidence and reconstructs the chain:



Diagnosis (auto-generated)

Severity CRITICAL · confidence 95% · dominant pattern

Layers implicated: 1 JVM · 2 Database · 3 Channel · 5 Destination · 7 Queue

Also caught: 'Lab Results Inbound' STOPPED and undetected

Ruled out: CPU saturation (71%), TLS faults (none in log)

Latent risks flagged: disk 2% free, file-descriptors 99%

Remediation (prioritized runbook)

NOW: restart stopped channel · relieve heap · isolate/drain the dead endpoint

DURABLE: migrate off Derby · configure pruning / retention

DURABLE: bounded retries + Rotate Queue · right-size heap

MONITOR: alert on stopped channels, queue depth, disk & error rates

SAFE ACTION offered: start_channel('Lab Results Inbound') — on approval

Three engines, one contract — resilient and deployable anywhere

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LLM Agent

Deepest reasoning over novel or ambiguous incidents.

Natural-language log understanding.

Best when connectivity & an API key are available.

Local Rule-Based

Deterministic, auditable, instant.

Runs fully OFFLINE — no API key, no network, no extra libraries.

Ideal for air-gapped hospital networks; no PHI leaves the host.

RAG Case-Memory

Learns from every resolved incident.

Surfaces “we’ve seen this before” with the past root cause & fix.

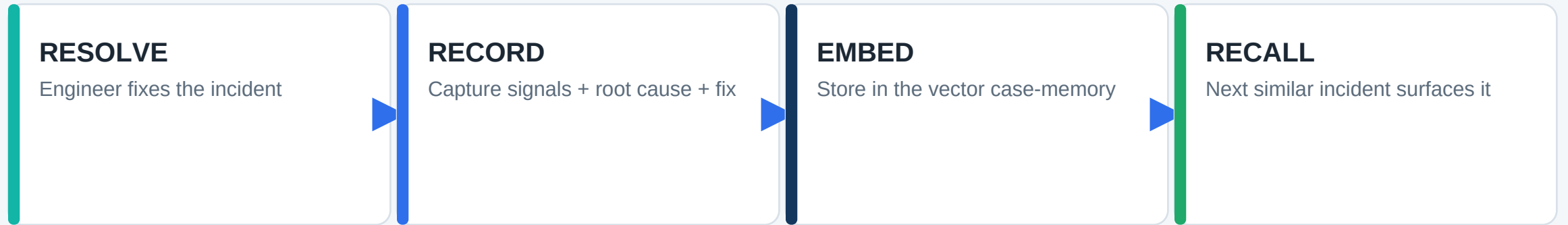
Grounds — never overrides — the verdict.

Why this matters to the enterprise

The same structured diagnosis is produced whether or not the cloud LLM is reachable — so the tool is dependable during the exact outage when it is needed most, and compliant where sending operational data off-site is prohibited.

It gets smarter with every incident

A closed feedback loop turns each resolved incident into institutional memory — no model retraining required.



Enterprise-friendly by default

Zero extra dependencies out of the box (deterministic embeddings).
Optional upgrade to semantic embeddings — one config change, no code change.
Runs on-prem; no incident data leaves your environment.

Trustworthy by design

Retrieval GROUND'S the diagnosis — attaches similar past cases and nudges confidence only on strong corroboration.
It never silently overrides a rule-based verdict, avoiding “confident-but-wrong” recall.

Safe, explainable, and auditable

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Human-in-the-loop

Safe actions run only after explicit confirmation — never autonomously.

Least privilege

Only two whitelisted actions (start / redeploy channel). Destructive steps are recommend-only.

Explainable

Every conclusion cites evidence, states what was ruled out, and carries a confidence score.

Data residency

Local & RAG modes keep all operational data and PHI on-host — no external calls.

Read-first

Investigation tools are strictly read-only; the engine cannot mutate config while diagnosing.

Verifiable quality

79 automated tests lock the behavior; the local engine is a deterministic golden baseline.

How we prove it works

The solution ships with a full offline test suite — no API key, no server — exercising every functional area.

79

unit tests, all passing

11/11

failure layers exercised

< 0.1s

full suite runtime

0

external dependencies

What the tests validate

Correct diagnosis

Critical multi-layer incident classified correctly.
“Doesn't cry wolf” — clean signals stay low severity.
Every detector checked at its threshold.

Resilience

Degrades gracefully when every sensor fails.
Handles missing / malformed engine data.
Agent tool-use loop verified without a network.

RAG integrity

Deterministic, normalized embeddings.
Recall + confidence-nudge behavior.
Survives an embedding-model swap.

What it changes for the customer

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↓ up to 70%

mean-time-to-resolve
on common incidents

24×7

senior-level triage
on every shift

↑ uptime

of critical HL7/FHIR
interfaces

↓ escalations

fewer 2 a.m. calls to
scarce specialists

Operational outcomes

Catch silent failures early — before clinicians and patients feel them.
Turn hours of manual log-hunting into a seconds-long, evidence-based diagnosis.
Standardize response quality — the newest engineer gets senior-grade guidance.
Prevention built in: each incident ends with the monitoring that stops the next one.

Strategic outcomes

Reduce dependence on a handful of irreplaceable integration experts.
Compliance-friendly: works on-prem / air-gapped with no PHI egress.
Institutional memory that compounds — knowledge stays even when staff turn over.
A defensible, auditable AI story for regulated healthcare environments.

Figures are illustrative targets based on the failure classes automated, not a contractual guarantee.

A day in the life — before vs. after

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BEFORE — reactive & manual

- A downstream EHR endpoint goes down overnight
- Queue silently grows to tens of thousands
- Heap fills; the engine freezes, then OOMs
- Morning: clinicians report missing results
- A specialist spends hours correlating logs
- Root cause found late; impact already felt



AFTER — proactive & AI-assisted

- Watcher spots the stopped channel / growing queue
- Agent auto-triages within seconds
- Causal chain + root cause surfaced immediately
- Similar past incident + proven fix recalled
- Safe action offered; on-call approves a restart
- Durable fixes + monitoring prevent recurrence

Flexible modes — from demo to production

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Triage

On-demand diagnosis of a single incident from a symptom or pasted logs.

Watch

Continuous polling; auto-triage the moment a channel stops or a queue crosses a threshold.

Feedback

Record a resolved incident into case-memory to strengthen future recall.

Composable flags — dial capability to the environment

--mock

Built-in realistic data — instant demo, no server.

--local

Rule-based engine — no API key, no network.

--rag

Case-memory grounding — learns over time.

--apply

Execute whitelisted safe actions with approval.

From a laptop demo to an on-prem production watcher, the same tool scales down and up — with the offline engine guaranteeing it always has a fallback.

Where we take it next

NEAR-TERM

- Semantic (vector) embeddings for sharper recall
- Richer log-narrative understanding
- Operator dashboard & shareable reports

MID-TERM

- More whitelisted safe actions (guarded)
- Multi-instance / fleet-wide triage
- Native alerting & ITSM / ticket integration

STRATEGIC

- Predictive detection before failure
- Auto-generated prevention & tuning advice
- Cross-customer anonymized pattern library

THE ASK

Turn scarce interface expertise into an always-on, self-improving AI resolver.

Pilot

Run against one production Mirth instance in read-only + local mode.

Measure

Baseline current MTTR; track reduction over a 60–90 day pilot.

Expand

Enable RAG case-memory and approved safe actions once trust is established.

Low risk, high leverage: read-only by default, offline-capable, fully tested —
and every incident makes it smarter.